knights-ST at AraSeg Shared Task 2026: Cross-Family Ensembling with Task-Adaptive Pretraining for Arabic Sentence Segmentation

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Abstract

We describe our submission to the AraSeg 2026 shared task on Arabic sentence segmentation (Paragraph-Aware, Closed track). We cast segmentation as binary token-boundary classification and build a system around three ideas: (i) task-adaptive pretraining (TAPT) of encoders on the task's own training text, (ii) regularized fine-tuning with R-Drop and boundary smoothing, and (iii) a hand-designed cross-family ensemble of architecturally diverse Arabic encoders. Our final system reaches a boundary F_1 of **94.8** on the blind test set (ranking 3rd of 12 teams, 0.5 points from the top-ranked team). Beyond the score, we report a practical finding backed by a series of blind submissions: on a small development set, a compact diversity-first ensemble *generalizes better* to unseen data than ensembles selected by exhaustively maximizing development F_1, and TAPT helps only *under-adapted* backbones while hurting a heavily-pretrained one. We also report five techniques that did *not* help, to save future practitioners the effort. Our code is publicly available at https://github.com/Body123/araseg-2026-knights-st.

1 Introduction

Sentence segmentation — deciding where one sentence ends and the next begins — is a prerequisite for most downstream Arabic NLP, yet Arabic offers no reliable punctuation shortcut: punctuation usage is inconsistent across genres and eras, and a large share of genuine sentence boundaries fall on plain words with no orthographic cue at all (Elkholy et al., 2026a). As we show in §5, boundaries *with* a punctuation or paragraph-break cue are comparatively easy, while boundaries on bare words require genuine syntactic and discourse-level disambiguation — and this difficulty is compounded by the **Closed track** constraint of AraSeg 2026, which permits pretrained language models but forbids training on any data beyond the officially released 174 training documents. Fine-tuning large encoders on data this small is itself a known source of instability, and it leaves little room to compensate for missing surface cues with brute-force data scale, which is the strategy that most improves word-boundary detection in the task's own scaling analysis (Elkholy et al., 2026a).

We approach this as binary token classification over sliding windows of a pretrained encoder (task details in §2.1), addressing the small-data/hard-boundary tension with three closed-track-legal ideas that add task signal without adding data: (i) continuing self-supervised pretraining on the training text itself before fine-tuning; (ii) regularizing fine-tuning so the small labeled set is used efficiently; and (iii) combining several architecturally different encoders so their errors partially cancel out. Our contributions are:

- A system combining TAPT (Gururangan et al., 2020), R-Drop (Liang et al., 2021), and boundary smoothing (Zhu and Li, 2022) over a cross-family ensemble, reaching **94.8** blind F_1 (3rd of 12 teams).
- An empirical analysis showing that (a) a small, hand-designed diverse ensemble transfers to blind data better than development-F_1-maximized selection, and (b) TAPT helps under-adapted backbones but degrades a heavily-pretrained one.
- A report of five techniques that did not improve results in our setting (Focal loss, CRF decoding, word-boundary loss reweighting, synthetic data augmentation, and TAPT on an already heavily-pretrained encoder).

2 Background

2.1 Shared Task Description

AraSeg 2026 poses Arabic sentence segmentation as token-level boundary prediction across four task

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 938–943
October 28-29, 2026 ©2026 Association for Computational Linguistics

variants that differ in whether paragraph breaks (Paragraph-Aware, PA, vs. not) and punctuation (vs. NoPnx) are available at inference time (Elkholy et al., 2026b). We participate in the **PA, Closed** track: paragraph breaks and punctuation are both available, and systems must be trained *exclusively* on the officially released AraSeg corpus, with no external labeled or unlabeled text. Systems are scored with boundary-level F_1, macro-averaged per document, on a development set (for model selection), a public test set, and a final blind test set released only in the evaluation phase.

2.2 Related Work

Sentence segmentation has largely been approached either with punctuation-driven rule-based tools or with sequence labeling. Punctuation-based segmenters (NLTK Punkt, spaCy's sentencizer, PySBD) achieve perfect precision but near-zero recall once punctuation is unreliable, which is common in Arabic. More recent work reframes segmentation as universal, punctuation-robust token classification: Segment Any Text (SaT; Frohmann et al., 2024) trains a compact multilingual model with noise-augmented pretraining and supports lightweight domain adaptation. For Arabic specifically, transformer-based sequence labeling has also been used for punctuation restoration (Sakr and Torki, 2023). We instead fine-tune several large pretrained Arabic encoders directly, favoring architectural diversity over a single universal model. Sequence labeling has classically used structured prediction, e.g., Conditional Random Fields (CRFs; Lafferty et al., 2001), which explicitly model transitions between adjacent labels — an assumption we find a poor fit for our task (§5): Arabic sentence boundaries rarely cluster, so a learned transition matrix mainly suppresses valid recall. For Arabic representation learning, we build on AraBERT (Antoun et al., 2020), CAMeLBERT (Inoue et al., 2021), ARBERT/MARBERT (Abdul-Mageed et al., 2021), and AraELECTRA (Antoun et al., 2021), adapting them further with task-adaptive pretraining (Gururangan et al., 2020) and regularizing with R-Drop (Liang et al., 2021) and boundary smoothing (Zhu and Li, 2022).

3 System Overview

Task and encoding. Each token receives a binary label $y \in \{0, 1\}$ (1 if a sentence boundary follows). Paragraph breaks are represented by a dedicated `[NEWLINE]` token added to the vocabulary. Documents longer than 512 sub-words are processed in overlapping windows and per-token logits are averaged across windows. We fine-tune a linear head on the encoder's final sub-word representation of each word, producing logits (z_0, z_1) per token, with class-weighted cross-entropy

$$\mathcal{L}_{\text{CE}} = -\big[w_1\, y \log p_1 + w_0(1-y)\log p_0\big], \quad (1)$$

where $p_c = \text{softmax}(z)_c$ and w_1=11, w_0=1, matching the corpus's ∼8% boundary rate. Figure 1 summarizes the pipeline.

Cross-family encoders. Ensemble gains come from *decorrelated* errors, so we deliberately span encoder families: AraBERT-large (Antoun et al., 2020), CAMeLBERT (MSA and Mix) (Inoue et al., 2021), ARBERTv2 (Abdul-Mageed et al., 2021), and AraELECTRA (Antoun et al., 2021), which uses a discriminative (replaced-token-detection) objective (Clark et al., 2020) rather than masked language modeling, making its errors especially complementary to the BERT-family (Devlin et al., 2019) models.

Task-adaptive pretraining. Before fine-tuning, we continue masked-language-model pretraining of each BERT-family encoder on the *training split's own text only*, following Gururangan et al. (2020). This is closed-track legal and adapts the encoder to the corpus's genres and the `[NEWLINE]` structure.

Regularized fine-tuning. R-Drop (Liang et al., 2021) runs two dropout-perturbed forward passes, obtaining two output distributions P_1, P_2 over $\{0, 1\}$ for the same input, and adds a symmetric KL penalty to Equation 1:

$$\mathcal{L}_{\text{R-Drop}} = \mathcal{L}_{\text{CE}} + \frac{\alpha}{2}\big[D_{\text{KL}}(P_1 \| P_2) + D_{\text{KL}}(P_2 \| P_1)\big], \quad (2)$$

with α=1.0. Boundary smoothing (Zhu and Li, 2022) instead replaces the hard target y in Equation 1 with a soft one: for every true boundary at position i, we set $\tilde{y}_i = 1-\epsilon$ and add $\epsilon/2$ to each adjacent word position (ϵ=0.1), countering boundary models' typical over-confidence; R-Drop and boundary smoothing combine by substituting $\tilde{y}$ for y in Equation 2.

Inference and ensembling. For a document with N ensemble members, we average class-1 logits,

$$\bar{z}_1 = \frac{1}{N}\sum_{i=1}^{N} z_1^{(i)}, \quad (3)$$

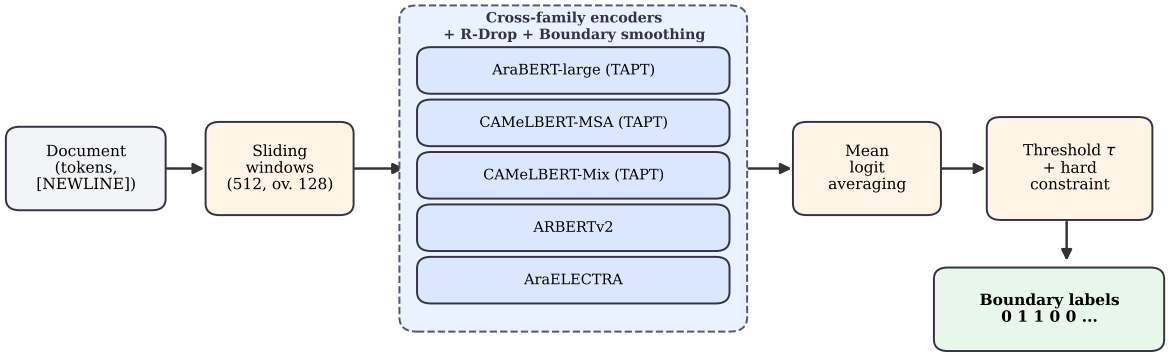


Figure 1: System overview. Each document is chunked into overlapping windows and encoded by an ensemble of architecturally diverse Arabic encoders; three are task-adaptively pretrained (TAPT) and all are fine-tuned with R-Drop and boundary smoothing. Per-token class-1 logits are averaged across models and windows, thresholded, and passed through a hard constraint that forces a boundary before every paragraph break (`[NEWLINE]`).

apply a single decision threshold τ tuned on development data ($\hat{y} = \mathbf{1}[\sigma(\bar{z}_1) > \tau]$), and enforce a hard constraint that forces a boundary before every paragraph break (always correct in this corpus).

Hand-designed ensemble. Our final system is a compact, *hand-designed* equal-weight ensemble of seven checkpoints spanning the four families in Figure 1 (three seeds of AraBERT-large, plus one checkpoint each of CAMeLBERT-Mix, CAMeLBERT-MSA, ARBERTv2, and AraELECTRA), chosen for diversity rather than by maximizing development F_1. §5 shows why this matters.

Configurations. We refer to a (backbone, TAPT, regularizer) combination as a *configuration*; Table 2 traces our main ensemble-building lineage from a CAMeLBERT-Mix baseline to the final cross-family ensemble. Separately, a single TAPT+R-Drop+boundary-smoothed AraBERT-large model reaches 94.1 test F_1 alone, our strongest non-ensembled configuration. We also trained, and report as negative results in §5, configurations adding a CRF layer, focal loss, word-boundary loss reweighting, synthetic data augmentation, and TAPT on ARBERTv2.

4 Experimental Setup

We use the AraSeg PA corpus (Elkholy et al., 2026b): 174 training, 222 development, and 262 public-test documents, plus a 100-document blind test set. Table 1 summarizes corpus statistics. Boundaries are heavily imbalanced (∼8% of tokens) and roughly a third fall on plain words with no punctuation or paragraph-break cue — these *word-based* boundaries are the hardest cases and, as we show in §5, the dominant source of remaining error. Over half of all documents exceed the 512-sub-word window, making sliding-window inference mandatory. The metric is boundary-level F_1, macro-averaged per document; all model selection uses development data, and the public-test and blind sets are scored once each. We fine-tune only on the training split, with 512-sub-word windows (overlap 128, or 256 for one member), AdamW at learning rate 2×10^{-5}, TAPT for 100 epochs with 15% masking, and three seeds per configuration.

	Train	Dev	Test
Documents	174	222	262
Tokens	128,173	163,609	158,626
Boundaries	10,657	12,985	12,509
Boundary rate (%)	8.3	7.9	7.9
Docs >512 tokens (%)	60	62	55
Boundary type breakdown (% of boundaries):			
pre-paragraph-break	35	37	38
punctuation-based	34	36	35
word-based (hard)	31	27	27

Table 1: AraSeg PA corpus statistics. Word-based boundaries carry no punctuation or paragraph-break cue and account for roughly a third of all boundaries.

5 Results

Main results. Table 2 traces each component's contribution: threshold tuning and the hard constraint give the single largest jump overall (+0.9), TAPT is the largest jump among the modeling techniques (+0.6), and cross-family diversity plus

System (public test)	F_1
CAMeLBERT-Mix, weighted CE	92.0
+ threshold tuning + hard constraint	92.9
+ 3-seed ensemble	93.5
+ cross-backbone (AraBERT-large)	93.8
+ TAPT	94.4
+ boundary smoothing	94.6
+ cross-family ensemble (final)	**94.7**

Table 2: Ablation: cumulative effect of each component on public-test F_1.

System	Dev	Test	Blind
Organizer baseline	—	92.8	93.9
Dev-F_1-maximized (large pool)	**95.3**	**94.8**	94.3
Hand-designed (final)	95.2	94.7	**94.8**

Table 3: Final systems. The dev-maximized ensemble wins on dev and public test yet *loses* on the blind set; the diversity-first design generalizes best.

boundary smoothing add the rest. Table 3 reports the final systems: our hand-designed ensemble scores **94.8** blind F_1, with near-identical public-test and blind scores (94.7 / 94.8) indicating strong generalization. In the official final evaluation, our system ranked **3rd of 12** teams, **0.5** F_1 points from the top-ranked team (ENSAR, 95.3) and 0.2 from 2nd place (DxMultiHead, 95.0).

Diversity beats exhaustive selection. Our strongest results came from continued experimentation after our first dev-phase submission (93.8 public-test F_1 for CAMeLBERT-Mix + AraBERT-large, officially confirmed, moving to 94.4 with TAPT and 94.6 with boundary smoothing). We then exhaustively searched ensemble compositions for the combination maximizing development F_1; across roughly a dozen blind submissions, the resulting higher-dev-F_1 candidates *consistently underperformed* our existing best on blind data. This dev-maximized ensemble scored highest on dev (95.3) and public test (94.8) but dropped to 94.3–94.4 on blind (Table 3), because selecting the argmax over a large pool overfits the 222-document development set. We therefore abandoned exhaustive selection for a compact, hand-designed, diversity-first ensemble which, though *lower* on dev and test, was best on blind (94.8): with a small dev set, prefer a compact, diverse design over aggressive selection — we view this trajectory itself, not only the final number, as one of our contributions.

Rank	Team	P	R	F_1
1	ENSAR	95.0	95.7	95.3
2	DxMultiHead	94.4	96.0	95.0
3	**knights-ST (ours)**	**94.8**	**95.1**	**94.8**
4	Al-Qahera	94.3	95.4	94.7
5	AUC DL Lab	95.1	94.2	94.5
6	NAMAA-Community	93.5	95.6	94.4
7	Itgan	94.6	94.7	94.4
8	iMak AI Lab	94.4	94.4	94.1
9	Baseline	95.5	92.7	93.9
10	Wathiq-SPU	93.2	94.0	93.4
11	Abjad AI	89.3	95.0	91.7
12	NLPBeasts	86.4	90.5	89.5

Table 4: Official final leaderboard for the AraSeg 2026 PA Closed track. Our system (knights-ST) ranked 3rd of 12 teams.

TAPT helps only under-adapted backbones. TAPT improved AraBERT-large and CAMeLBERT but *degraded* ARBERTv2 (test $94.0 \rightarrow 93.7$): ARBERTv2's very large pretraining corpus means continued MLM on only 174 documents induces mild forgetting rather than useful adaptation, so TAPT helps most when the base encoder is comparatively under-adapted.

Error analysis: where errors remain. On development data, 70% of missed boundaries fall on punctuation, dominated by the comma, which is *also* our largest source of false positives: the tokens we miss are the tokens we over-predict. At matched recall (95.7, our system at τ=0.55) against the top-ranked final submission, its precision (95.0) exceeds ours (94.1) by 0.9 points — suggesting a model-quality gap in disambiguating ambiguous punctuation, not a decision-threshold artifact.

What did not work. Five techniques did not improve results in our setting. Focal loss underperformed weighted cross-entropy, already well-suited to our fixed imbalance ratio. A CRF layer (Lafferty et al., 2001) lowered recall, since Arabic boundaries rarely occur on consecutive tokens and the learned transition matrix mainly penalizes valid predictions. Word-boundary loss reweighting gave inconsistent, backbone-dependent changes; synthetic data from recombined training sentences introduced unnatural transitions that hurt more than the extra examples helped; and TAPT hurt ARBERTv2 for the reason discussed above.

6 Conclusion

We presented a cross-family Arabic sentence-segmentation ensemble combining task-adaptive

pretraining, R-Drop, and boundary smoothing, reaching 94.8 blind F_1 (3rd of 12 teams). Our two main findings — diversity-first ensembling generalizes better than exhaustive dev-set selection, and TAPT helps only under-adapted backbones — together with five negative results, should transfer to other low-resource settings. The remaining gap to the top result is concentrated in genuinely ambiguous punctuation (chiefly the comma), not decision thresholds or ensembling choices, suggesting future work should target finer-grained punctuation disambiguation over further architecture search. Our study covers only MSA PA data with a development set small enough to be overfit by aggressive search; the NoPnx/NP conditions, dialectal text, and a theoretically grounded alternative to exhaustive ensemble selection remain open.

Limitations

Our study covers only Modern Standard Arabic in the Paragraph-Aware condition; the NoPnx (no punctuation) and NP (no paragraph breaks) task variants and dialectal Arabic text remain untested. Since our training corpus is MSA-heavy, our segmenter likely faces the same register-shift degradation seen in other MSA-trained Arabic NLP systems; a per-sentence Arabic dialectness estimate (Sakr et al., 2024) could flag such inputs or guide a dialect-adaptive extension of our approach, which we leave to future work. Task-adaptive pretraining and R-Drop roughly double training cost relative to plain fine-tuning (continued masked-language-model pretraining, plus two forward passes per step). Our ensemble design and decision threshold were selected using a 222-document development set, which we show (§5) is small enough to be overfit by aggressive search; our diversity-first remedy is an empirical finding rather than a theoretically guaranteed one, and a principled alternative to exhaustive ensemble selection remains open. Finally, our own error analysis shows the residual errors are concentrated in genuinely ambiguous punctuation (chiefly the comma, which is simultaneously our top false-negative and false-positive class), consistent with the comma being the hardest of six punctuation classes (lowest F1) in independent Arabic punctuation-restoration results (Sakr and Torki, 2023).

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